

1.P-N JUNCTION DIODE CHARACTERISTICS

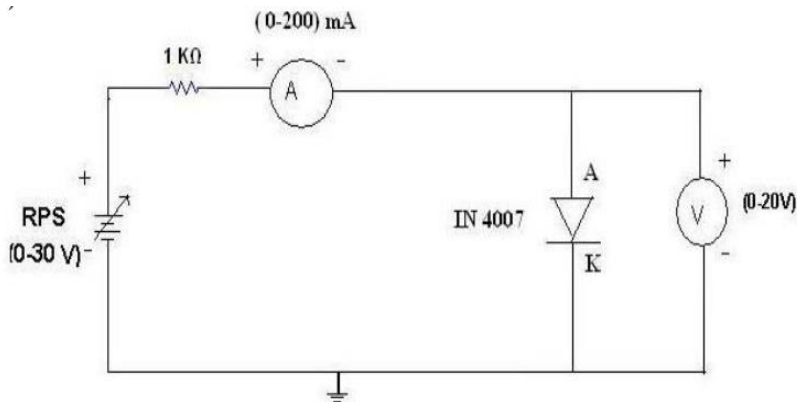
AIM: 1. To observe and draw the Forward and Reverse bias V-I Characteristics of a P-N Junction diode.
2. To calculate static and dynamic resistance in both forward and Reverse Bias Condition.

APPARATUS:

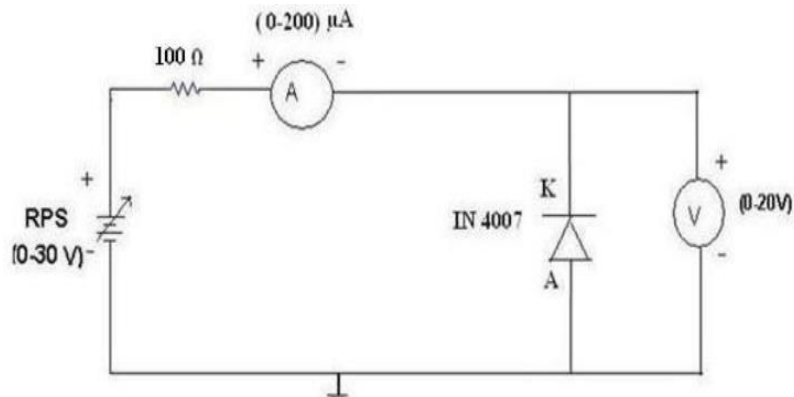
1. P-N Diode IN4007 - 1No.
2. Regulated Power supply (0-30V) - 1No.
3. Resistor $1\text{K}\Omega$ - 1No.
4. Ammeter (0-20 mA) - 1No
5. Ammeter (0-200 μA) - 1No.
6. Voltmeter (0-20V) - 2No.
7. Bread board
8. Connecting wires

CIRCUIT DIAGRAM:

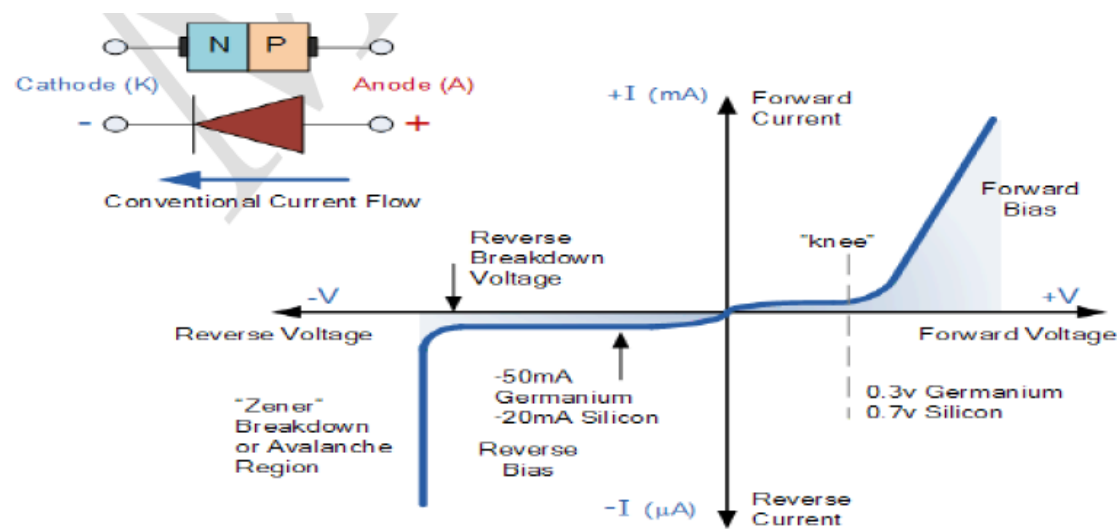
A. Forward bias:



B.Reverse Bias:



MODEL GRAPH:



OBSERVATIONS:

A) FORWARD BIAS:

S.NO	Applied Voltage(V)	Forward Voltage(Vf)	Forward Current (If (mA))

B) REVERSE BIAS:

S.NO	Applied Voltage(V)	Reverse Voltage(VR)	Reverse Current(IR(μA))

OBSERVATIONS:

Calcutions:

Calculation of Static and Dynamic Resistance for a given diode.

In forward bias condition;

Static Resistance , $R_s = V_f / I_f =$

Dynamic Resistance, $R_D = \Delta V_f / \Delta I_f =$

In Reverse bias condition:

Static Resistance , $R_s = V_R / I_R =$

Dynamic Resistance, $R_D = \Delta V_R / \Delta I_R =$

PROCEDURE:

A) FORWARD BIAS:

1. Connections are made as per the circuit diagram.
2. for forward bias, the RPS +ve is connected to the anode of the diode and RPS –ve is connected to the cathode of the diode
3. Switch on the power supply and increases the input voltage (supply voltage) in Steps of 0.1V
4. Note down the corresponding current flowing through the diode and voltage across the diode for each and every step of the input voltage.
5. The reading of voltage and current are tabulated.
6. Graph is plotted between voltage (V_f) on X-axis and current (I_f) on Y-axis.

B) REVERSE BIAS:

1. Connections are made as per the circuit diagram
2. For reverse bias, the RPS +ve is connected to the cathode of the diode and RPS –ve is connected to the anode of the diode.
3. Switch on the power supply and increase the input voltage (supply voltage) in Steps of 1V.
4. Note down the corresponding current flowing through the diode voltage across the diode for each and every step of the input voltage.
5. The readings of voltage and current are tabulated
6. Graph is plotted between voltage (V_R) on X-axis and current (I_R) on Y-axis.

PRECAUTIONS:

1. All the connections should be correct.
2. Parallax error should be avoided while taking the readings from the Analog meters

RESULT:

2.ZENER DIODE CHARACTERISTICS AND ZENER AS VOLTAGE REGULATOR

AIM:

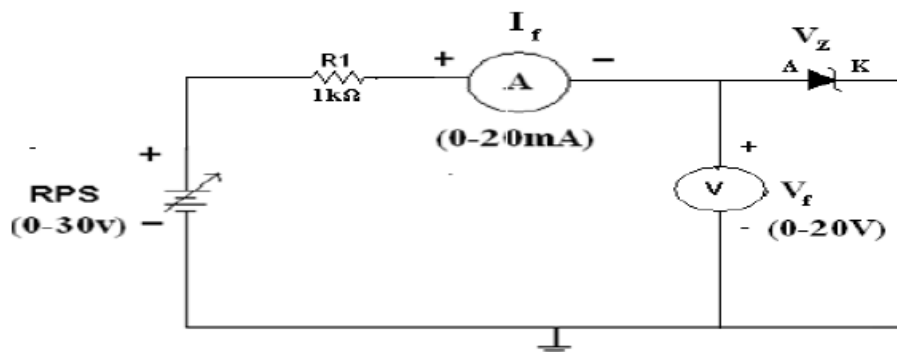
- a) To observe and draw the static characteristics of a zener diode
- b) To find the voltage regulation of a given zener diode

APPARATUS:

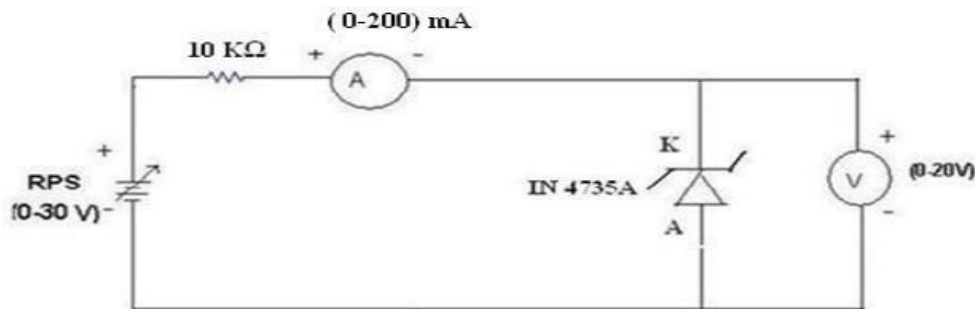
1. Zener diode - 1No.
2. Regulated Power Supply (0-30v) - 1No.
3. Voltmeter (0-20v) - 1No.
4. Ammeter (0-20mA) - 1No.
5. Resistor (1K ohm)
6. Bread Board
7. Connecting wires

CIRCUIT DIAGRAM

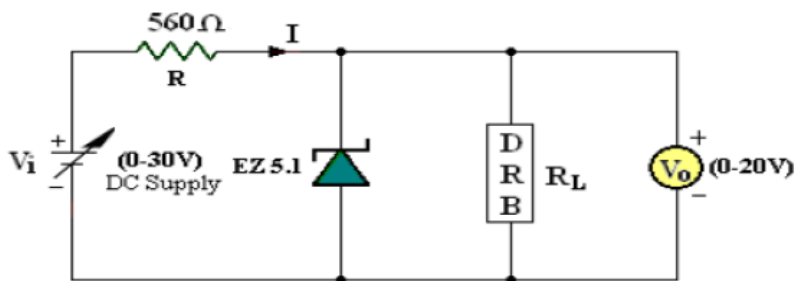
A) STATIC CHARACTERISTICS :



b) REVERSE BIAS CHARACTERISTICS:



VOLTAGE REGULATION:

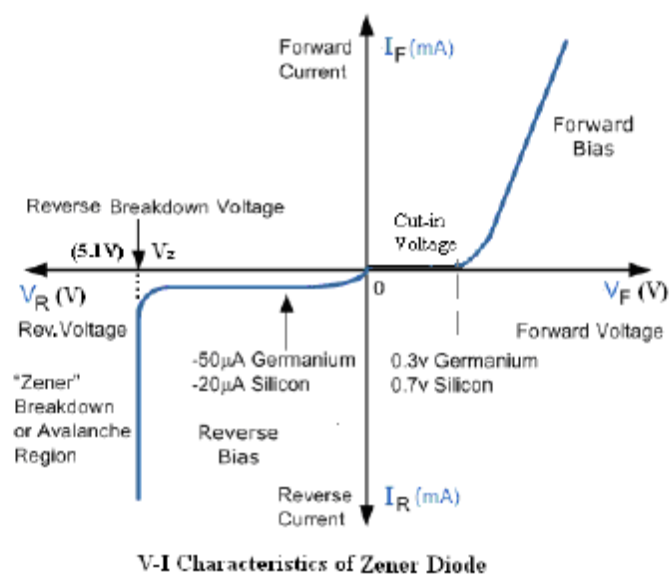


Zener Diode Voltage Regulation

Zener Diode Voltage Regulation

MODEL GRAPHS:

ZENER DIODE CHARACTERISTICS:



OBSERVATIONS:

A) FORWARD BIAS CHARACTERISTICS :

S.NO	Applied Voltage(V)	Forward Voltage(Vf)	Forward Current(Ir(mA))
1.			
2.			
3.			

B) REVERSE BIAS CHARACTERISTICS:

S.NO	Applied Voltage(V)	Reverse Voltage(VR)	Reverse Current(IR(mA))
1.			
2.			
3.			

PROCEDURE:**A) Static characteristics:**

1. Connections are made as per the circuit diagram.
2. The Regulated power supply voltage is increased in steps.
3. The Forward current (I_f), and the forward voltage (V_f) are observed and then noted in the tabular form.
4. A graph is plotted between Forward current (I_f) on X-axis and the forward voltage (V_f) on Y-axis.

B) Load Regulation characteristics:

1. Connect the Circuit as per the Circuit Diagram on the bread board.
2. By changing the load Resistance, kept constant I/P Voltage at 5V, 10 V, 15 V as per table given below. Take the readings of O/P Voltmeter ($V_o=V_z$).
3. Now by changing the I/P Voltage, kept constant load Resistance at 1K, 2K, 3K as per table given below. Take the readings of O/P Voltmeter ($V_o=V_z$).

S.No	$R_L (\Omega)$	$V_{i1}=5V$ $V_o (V)$	$V_{i2}=10V$ $V_o (V)$	$V_{i3}=15V$ $V_o (V)$
1	100			
2	300			
3	500			
4	700			
5	900			
6	1K			
7	3K			
8	5K			
9	7K			
10	10K			

$V_i (V)$	$R_{L1}=1K\Omega$ $V_o (V)$	$R_{L2}=2K\Omega$ $V_o (V)$	$R_{L3}=3K\Omega$ $V_o (V)$
0			
1			
3			
5			
7			
9			
11			
13			
15			
20			

PRECAUTIONS:

1. The terminals of the zener diode should be properly identified
2. While determined the load regulation, load should not be immediately shorted.
3. Should be ensured that the applied voltages & currents do not exceed the ratings of the diode.

RESULT:

3. HALF -WAVE RECTIFIER WITH AND WITHOUT FILTER

AIM: To examine the input and output waveforms of half wave Rectifier and also Calculate its load regulation and ripple factor.

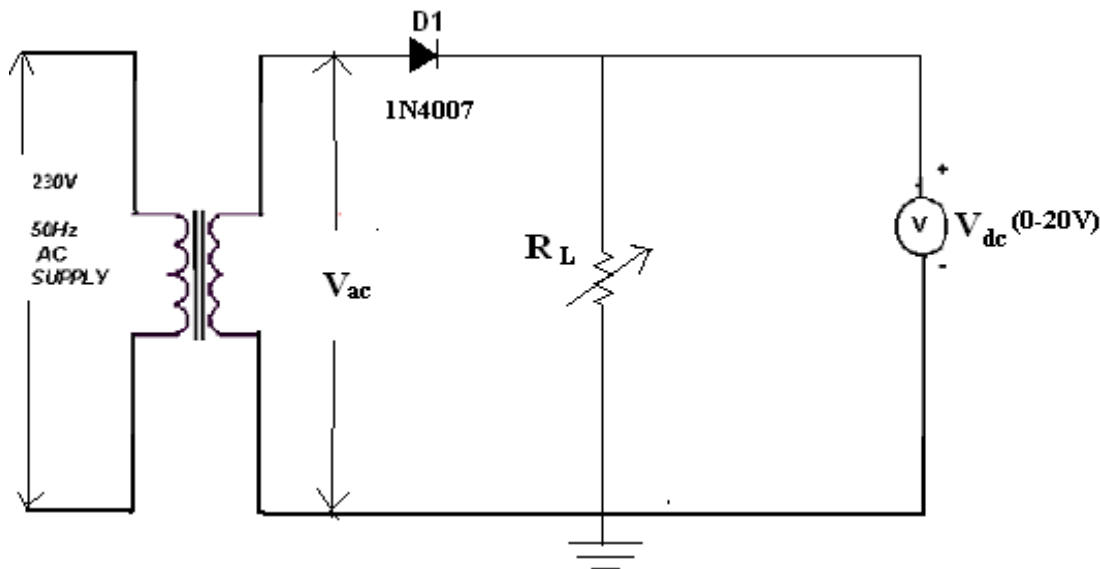
1. With Filter
2. Without Filter

APPARATUS:

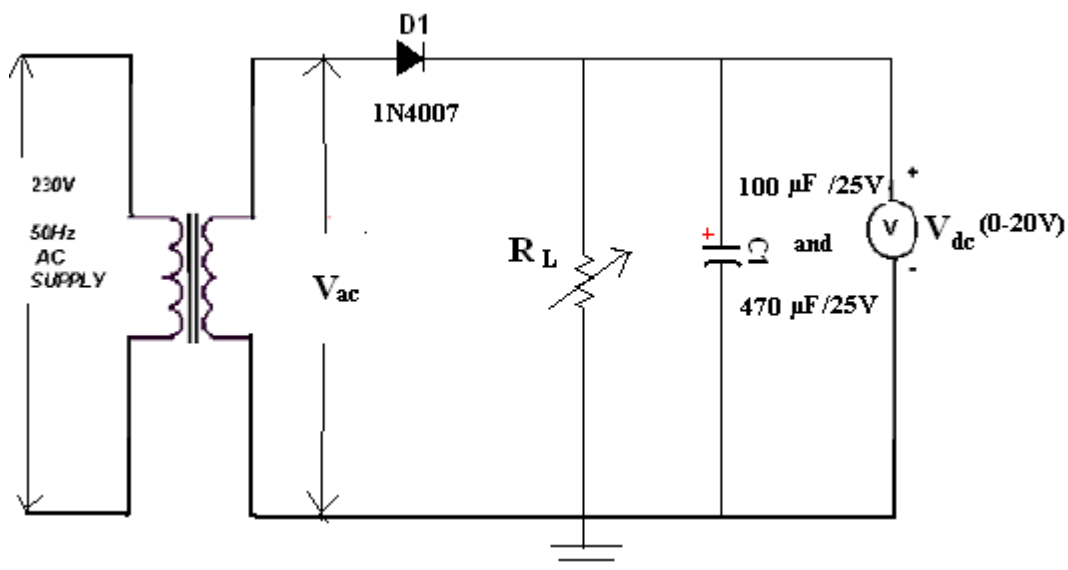
Digital Multimeter - 1No.
Transformer (6V-0-6V) - 1No.
Diode, 1N4007 - 1No.
Capacitor 100 μ f/470 μ f - 1No.
Decade Resistance Box - 1No. Breadboard
CRO and CRO probes Connecting wires

CIRCUIT DIAGRAM:

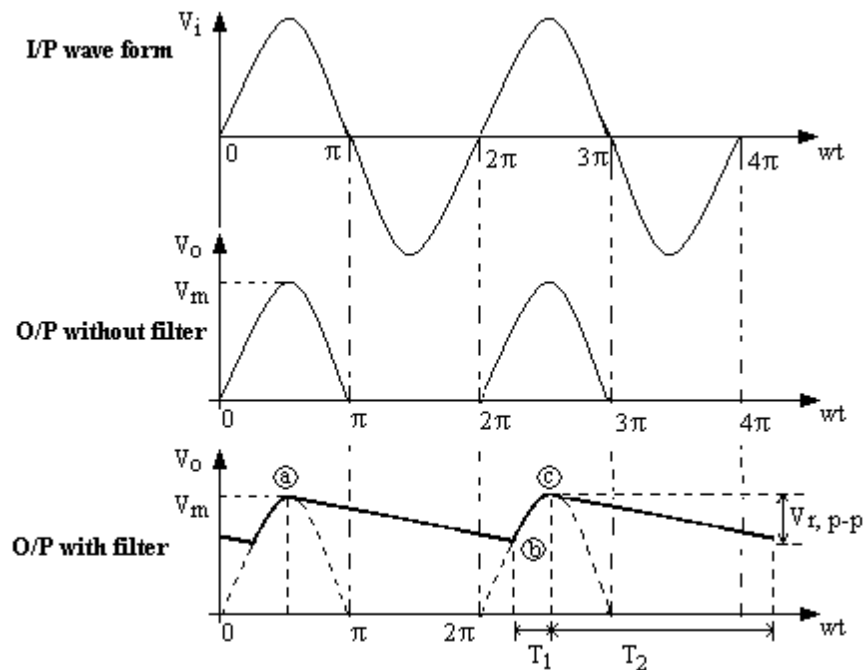
A) Half Wave Rectifier Without Filter:



C) Half Wave Rectifier With Filter



**MODEL WAVEFORMS:
A WAVEFORMS:**



Half-wave Rectifier with capacitor filter wave form

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.
3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output of the rectifier.
4. Find the theoretical value of dc voltage by using the formula, $V_{dc} = V_m / \pi$
Where, $V_m = 2V_{rms}$, (V_{rms} =output ac voltage.)
5. The Ripple factor is calculated by using the formula
 $r = \text{ac output voltage} / \text{dc output voltage}$

WITHOUT FILTER:

V no load Voltage (Vdc) = V

S.No	Load Resistance R_L kilo-ohm S	O/P Voltage (Vo)		Ripple factor $\left(\gamma = \frac{V_{ac}}{V_{dc}}\right)$	% of Regulation $\left(\frac{V_{NL} - V_{FL}}{V_{NL}} * 100\%\right)$
		V _{ac} (V)	V _{dc} (V)		
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				
7	7				
8	8				

WITH CAPACITOR FILTER:

V no load Voltage (Vdc) = V

S.No	Load Resistance R_L kilo-ohm S	O/P Voltage (Vo)		Ripple factor $\left(\gamma = \frac{V_{ac}}{V_{dc}}\right)$	% of Regulation $\left(\frac{V_{NL} - V_{FL}}{V_{NL}} * 100\%\right)$
		V _{ac} (V)	V _{dc} (V)		
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				
7	7				
8	8				

REGULATION CHARACTERISTICS:

1. Connections are made as per the circuit diagram.
2. By increasing the value of the rheostat, the voltage across the load and current flowing through the load are measured.
3. The reading is tabulated.
4. From the value of no-load voltages, the %regulation is calculated using the formula,

$$\% \text{Regulation} = [(V_{NL} - V_{FL}) / V_{FL}] * 100$$

PRECAUTIONS:

1. The primary and secondary side of the transformer should be carefully identified
2. The polarities of all the diodes should be carefully identified.
3. While determining the % regulation, first full load should be applied and then it should be decremented in steps

RESULT:

1. FULL-WAVE RECTIFIER WITH AND WITHOUT FILTER

AIM: To Examine the input and output waveforms of Full Wave Rectifier and also calculate its load regulation and ripple factor

1. With Filter
2. Without Filter

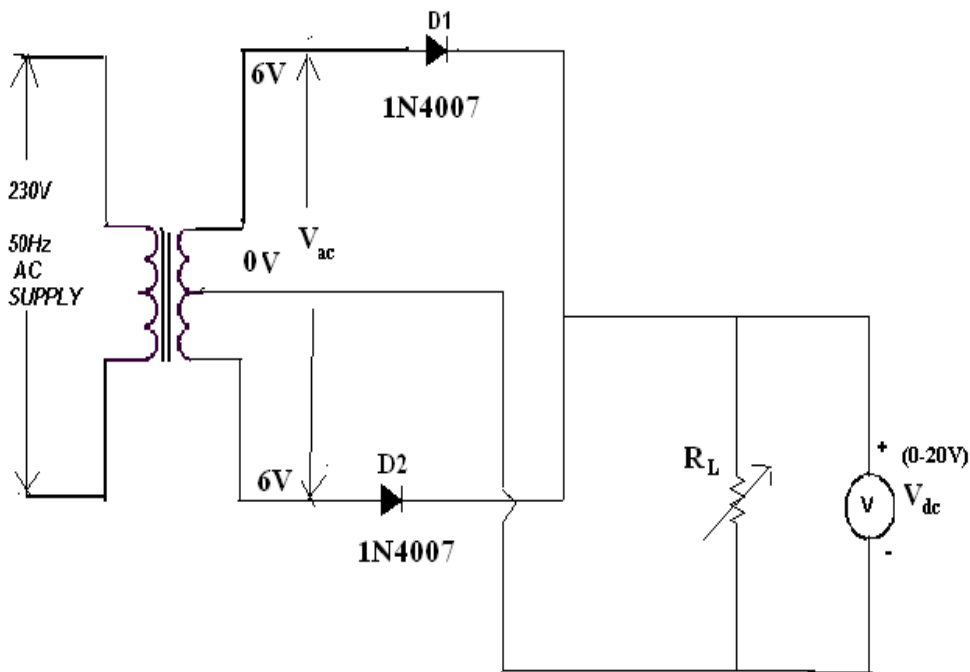
APPARATUS:

Digital multimeters Multimeter - 1No. Transformer (6V-0-6V) - 1No.
Diode, 1N4007 - 2No.
Capacitor $100\mu\text{f}/470\mu\text{f}$ - 1No.
Decade Resistance Box - 1No. Breadboard
CRO and CRO probes Connecting wires

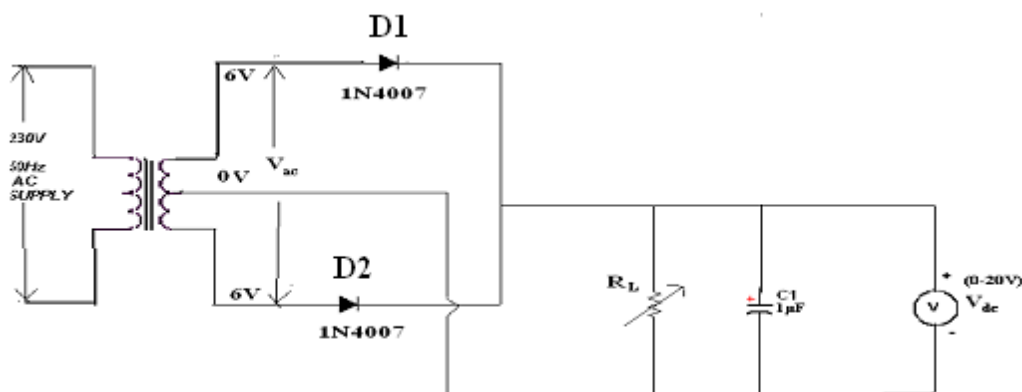
CIRCUIT DIAGRAM

A) FULL WAVE RECTIFIER WITHOUT FILTER:

B) FULL WAVE RECTIFIER WITH FILTER:

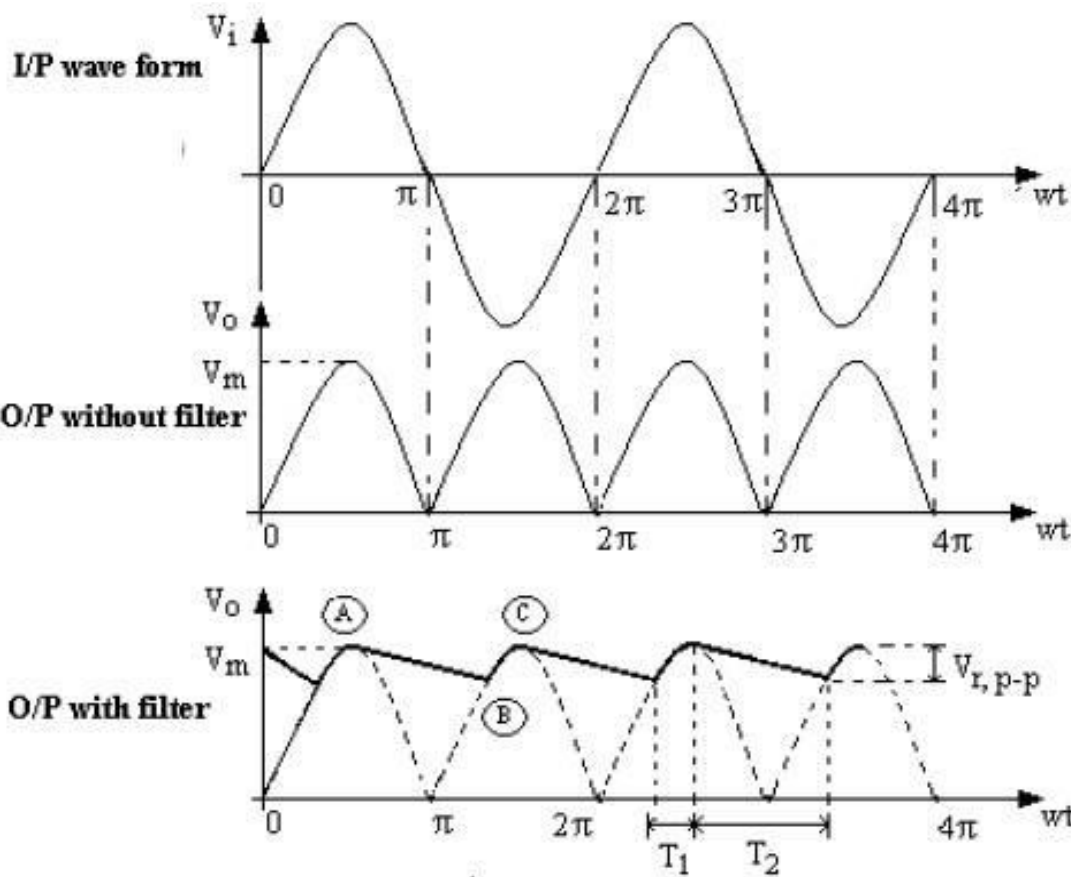


B) FULL WAVE RECTIFIER WITH FILTER:



MODEL WAVEFORMS:

A.WAVEFORMS:



Full-wave Rectifier with capacitor filter wave form

WITHOUT FILTER:

V no load Voltage (Vdc) = V

S.No	Load Resistance R_L kilo-ohm	O/P Voltage (Vo)		Ripple factor $\left(\gamma = \frac{V_{ac}}{V_{dc}}\right)$	% of Regulation $\left(\frac{V_{NL} - V_{FL}}{V_{NL}} * 100\%\right)$
		V _{ac} (V)	V _{dc} (V)		
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				
7	7				
8	8				

WITH CAPACITOR FILTER:

V no load Voltage (Vdc) = V

S.No	Load Resistance R_L kilo-ohm	O/P Voltage (Vo)		Ripple factor $\left(\gamma = \frac{V_{ac}}{V_{dc}}\right)$	% of Regulation $\left(\frac{V_{NL} - V_{FL}}{V_{NL}} * 100\%\right)$
		V _{ac} (V)	V _{dc} (V)		
1	1				
2	2				
3	3				
4	4				
5	5				
6	6				
7	7				
8	8				

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the ac mains to the primary side of the transformer and the secondary side to the rectifier.
3. Measure the ac voltage at the input side of the rectifier.
4. Measure both ac and dc voltages at the output side the rectifier.
5. Find the theoretical value of the dc voltage by using the formula $V_{dc} = 2V_m/\pi$
6. Connect the filter capacitor across the load resistor and measure the values of V_{ac} and V_{dc} at the output
7. The theoretical values of Ripple factors with and without capacitor are calculated.
8. From the values of V_{ac} and V_{dc} practical values of Ripple factors are calculated. The practical values are compared with theoretical values.

PRECAUTIONS:

1. The primary and secondary side of the transformer should be carefully identified.
2. The polarities of all the diodes should be carefully identified.

RESULT:

5. INPUT AND OUTPUT CHARACTERISTICS OF TRANSISTOR CB CONFIGURATION

AIM:

1. To observe and draw the input and output characteristics of a transistor connected in common base configuration.
2. To find α of the given transistor and also its input and output Resistances.

APPARATUS:

Transistor, BC107 -1No.

Regulated power supply (0-30V) -1No.

Voltmeter (0-20V) - 2No.

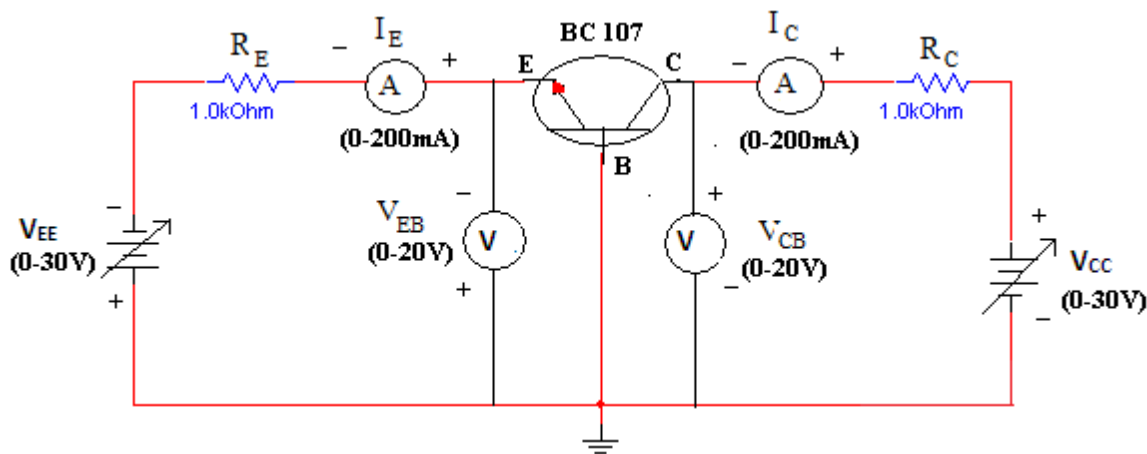
Ammeters (0-10mA) - 2No.

Resistor, $1\text{k}\Omega$ - 2No

Bread board

Connecting wires

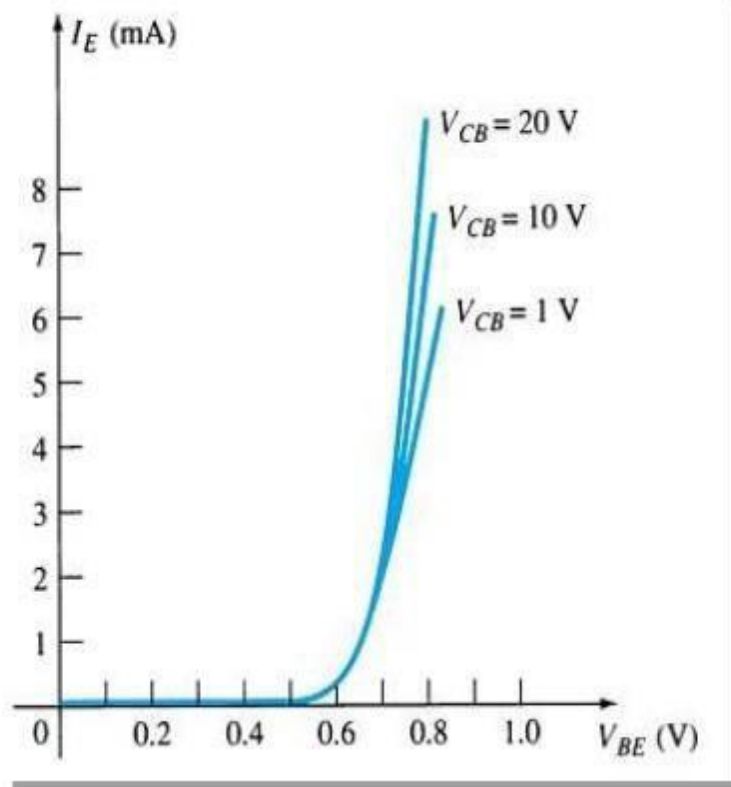
CIRCUIT DIAGRAM:



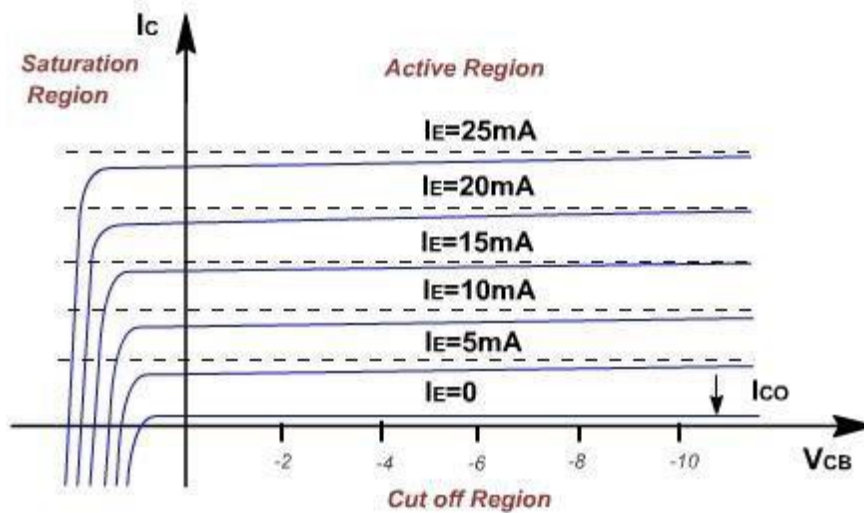
MODEL GRAPHS:

A) INPUT CHARACTERISTICS:

Input Characteristics



B) OUTPUT CHARACTERISTICS



OBSERVATIONS:**A) INPUT CHARACTERISTICS:**

SN.	V _{EE} (V)	V _{CB} =1V		V _{CB} = 2V		V _{CB} = 4V	
		V _{EB} (V)	I _E (mA)	V _{EB} (V)	I _E (mA)	V _{EB} (V)	I _E (mA)
1.							
2.							
3.							
4.							

B) OUTPUT CHARACTERISTICS:

SN.	V _{cc} (V)	I _E =10mA		I _E =20mA		I _E =30mA	
		V _{CB} (V)	I _C (mA)	V _{CB} (V)	I _C (mA)	V _{CB} (V)	I _C (mA)
1.							
2.							
3.							
4.							

PROCEDURE:**A) INPUT CHARACTERISTICS:**

1. Connections are made as per the circuit diagram.
2. For plotting the input characteristics, the output voltage V_{CE} is kept constant at 0V and for different values of V_{EE} note down the values of I_E and V_{BE}
3. Repeat the above step keeping V_{CB} at 2V, 4V, and 6V and all the readings are tabulated.
4. A graph is drawn between V_{EB} and I_E for constant V_{CB}.

B) OUTPUT CHARACTERISTICS:

1. Connections are made as per the circuit diagram.
2. For plotting the output characteristics, the input I_E is kept constant at 0.5mA and for different values of V_{CC}, note down the values of I_C and V_{CB}.
3. Repeat the above step for the values of I_E at 1mA, 5mA and all the readings are tabulated.
4. A graph is drawn between V_{CB} and I_C for constant I_E

PRECAUTIONS:

1. The supply voltages should not exceed the rating of the transistor.
2. Meters should be connected properly according to their polarities.

RESULT:

6. INPUT AND OUTPUT CHARACTERISTICS OF TRANSISTOR CE CONFIGURATION

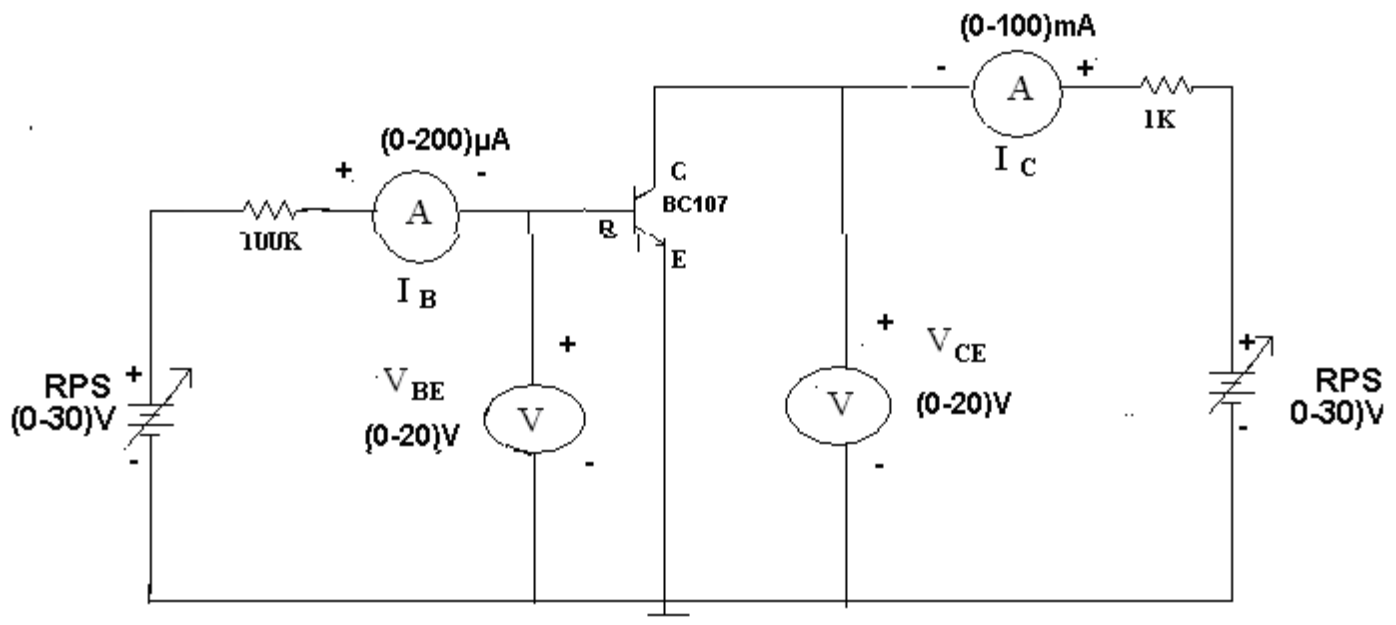
AIM:

1. To draw the input and output characteristics of transistor connected in CE configuration
2. To find β of the given transistor and also its input and output Resistances

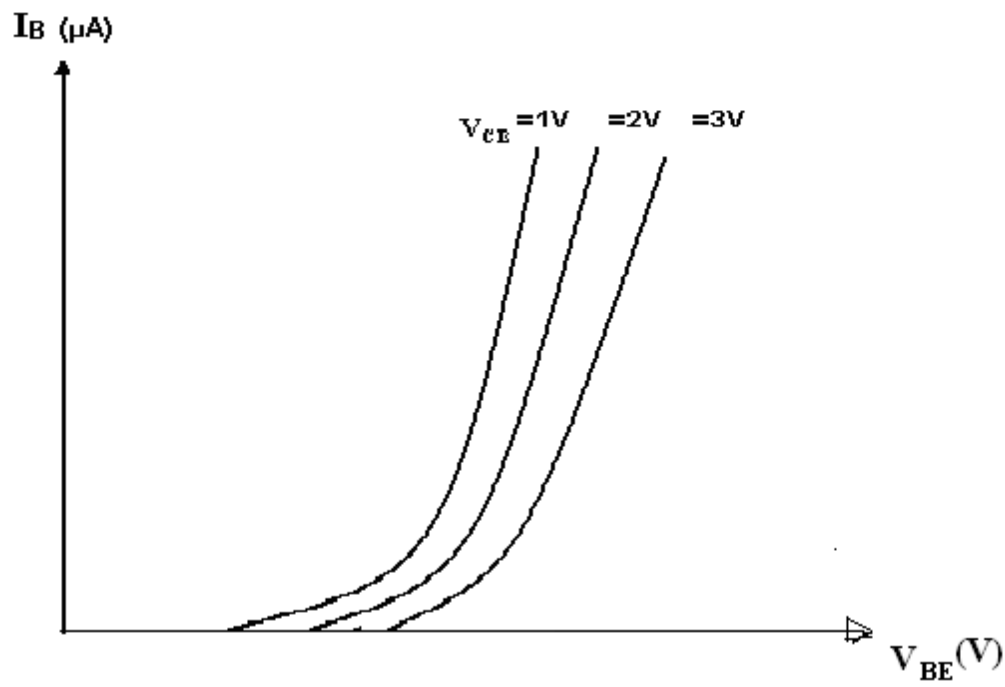
APPARATUS:

Transistor, BC107 -1No.
Regulated power supply (0-30V) -1No.
Voltmeter (0-20V) - 2No.
Ammeters (0-20mA) -1No.
Ammeters (0-200 μ A) -1No.
Resistor, 100 Ω -1No
Resistor, 1K Ω -1No.
Bread board
Connecting wires

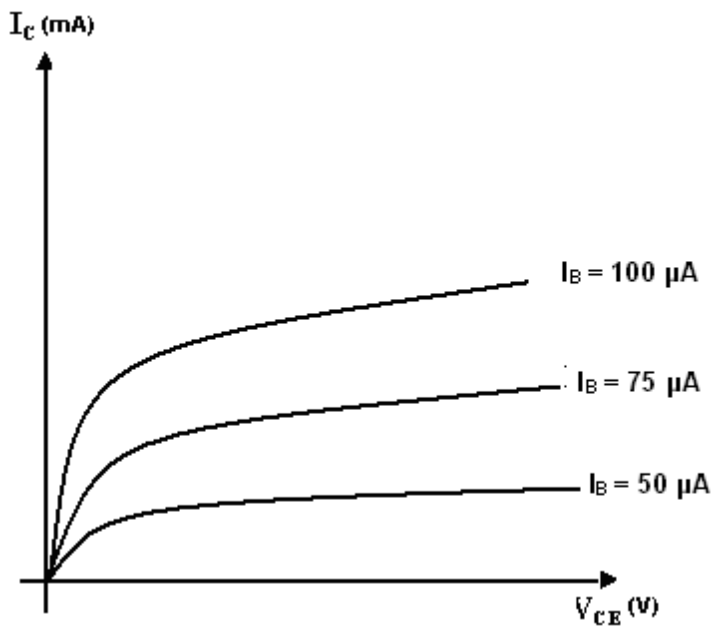
CIRCUIT DIAGRAM:



MODEL GRAPHS:
A) INPUT CHARACTERISTICS:



B) OUTPUT CHARACTERISTICS:



OBSERVATIONS:
A) INPUT CHARACTERISTICS:

SN.	V _{BB}	V _{CE} = 1V		V _{CE} = 2V		V _{CE} = 4V	
		V _{BE} (V)	I _B (μA)	V _{BE} (V)	I _B (μA)	V _{BE} (V)	I _B (μA)
1.							
2.							
3.							

B) OUTPUT CHARACTERISTICS:

SN.	$I_B = 50 \mu A$		$I_B = 570 \mu A$		$I_B = 100 \mu A$	
	$V_{CE}(V)$	$I_C(mA)$	$V_{CE}(V)$	$I_C(mA)$	$V_{CE}(V)$	$I_C(mA)$
1.						
2.						
3.						

PROCEDURE:**A) INPUT CHARACTERISTICS:**

1. Connect the circuit as per the circuit diagram.
2. For plotting the input characteristics the output voltage V_{CE} is kept constant at 1V and for different values of V_{BB} , note down the values of I_B and V_{BE}
3. Repeat the above step by keeping V_{CE} at 2V and 4V and tabulate all the readings.
4. plot the graph between V_{BE} and I_B for constant V_{CE}

B) OUTPUT CHARACTERISTICS:

1. Connect the circuit as per the circuit diagram.
2. for plotting the output characteristics the input current I_B is kept constant at $50 \mu A$ and for different values of V_{CC} note down the values of I_C and V_{CE} .
3. Repeat the above step by keeping I_B at $75 \mu A$ and $100 \mu A$ and tabulate the all the Readings.
4. plot the graph between V_{CE} and I_C for constant I_B .

PRECAUTIONS:

1. The supply voltage should not exceed the rating of the transistor.
2. Meters should be connected properly according to their polarities.

RESULT:

